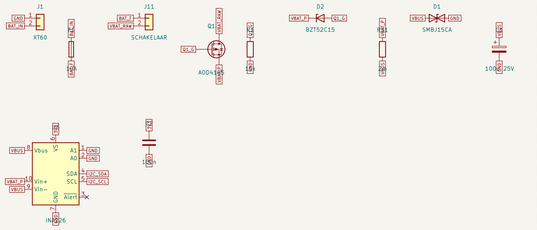
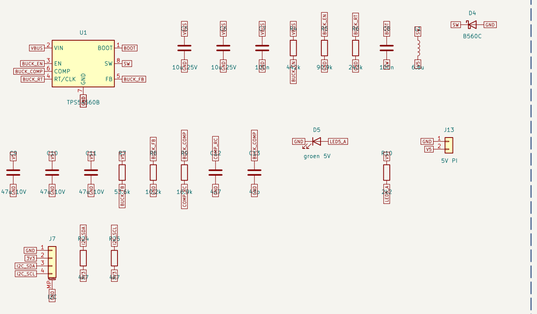
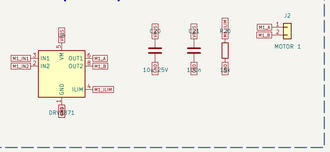
BATTERIJ 3S: XT60, ZEKERING, SCHAKELAAR, STROOMMETING



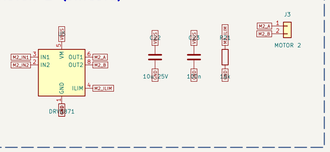
BUCK 5 V 5 A (PI 4 + SERVO)



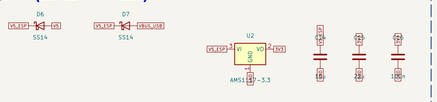
MOTOR 1 (DRV8871)



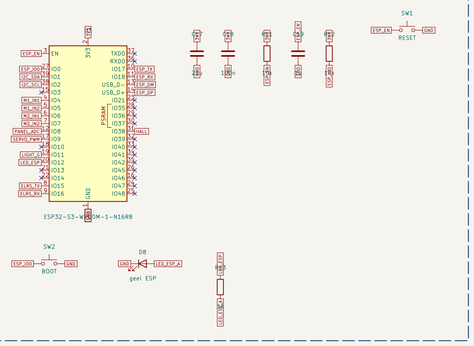
MOTOR 2 (DRV8871)



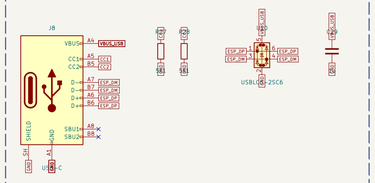
3,3 V (ESP32 + INA226)



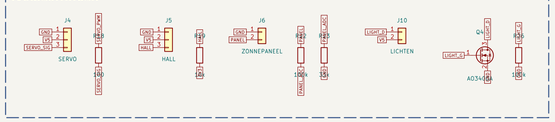
ESP32-S3



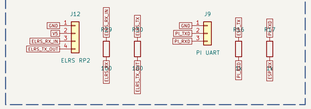
USB-C (FLASHEN / DEBUG VAN BUITENAF)



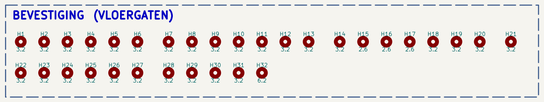
AANSLUITINGEN



ELRS-ONTVANGER + PI UART



BEVESTIGING (VLOERGATEN)



SC-02 FLOOR BOARD v2.0 — the whole car floor is this PCB (2 mm, 4 layers): ESP32-S3 + 2x DRV8871 + INA226 + 5 V 5 A buck (generated by tools/build_sch.py from tools/design.py)
Battery: XT60 -> 10 A fuse -> power switch (pads, switch in the floor hole) -> P-MOSFET reverse protection -> 2 mOhm shunt (INA226 0x40) -> VBUS: motors + buck.
Motors: DRV8871 IN1/IN2 (M1: IO4/IO5, M2: IO6/IO7), current limit 64 / 18k = 3.55 A. Servo IO9 (100 R), hall A3144 IO38 (10k pull-up), lights IO11 (AO3400A), panel ADC IO8 (100k/33k).
Pi 4: UART pads (ESP IO17 TX / IO18 RX via 1k) to Pi pins 8/10 + 5V PI pads (pin 2/4 + 6). USB-C at the floor edge (body opening) for flashing. ELRS receiver (RadioMaster RP2, CRSF 420k) on IO15 TX / IO16 RX.

whole car floor = PCB: ESP32-S3 + 2x DRV8871 + INA226 + buck
Kobe Van Robaeys
Sheet: /
File: sc02_drive.kicad_sch
Title: SC-02 floor board
Size: A1 | Date: 2026-10-18 | **Rev: 2.0**
KiCad E.D.A. 10.0.3 | Id: 1/1